By how much does dietary salt reduction lower blood pressure? III--Analysis of data from trials of salt reduction.

OBJECTIVE: To determine whether the reduction in blood pressure achieved in trials of dietary salt reduction is quantitatively consistent with estimates derived from blood pressure and sodium intake in different populations, and, if so, to estimate the impact of reducing dietary salt on mortality from stroke and ischaemic heart disease. DESIGN: Analysis of the results of 68 crossover trials and 10 randomised controlled trials of dietary salt reduction. MAIN OUTCOME MEASURE: Comparison of observed reductions in systolic blood pressure for each trial with predicted values calculated from between population analysis. RESULTS: In the 45 trials in which salt reduction lasted four weeks or less the observed reductions in blood pressure were less than those predicted, with the difference between observed and predicted reductions being greatest in the trials of shortest duration. In the 33 trials lasting five weeks or longer the predicted reductions in individual trials closely matched a wide range of observed reductions. This applied for all age groups and for people with both high and normal levels of blood pressure. In people aged 50-59 years a reduction in daily sodium intake of 50 mmol (about 3 g of salt), attainable by moderate dietary salt reduction would, after a few weeks, lower systolic blood pressure by an average of 5 mm Hg, and by 7 mm Hg in those with high blood pressure (170 mm Hg); diastolic blood pressure would be lowered by about half as much. It is estimated that such a reduction in salt intake by a whole Western population would reduce the incidence of stroke by 22% and of ischaemic heart disease by 16% [corrected]. CONCLUSIONS: The results from the trials support the estimates from the observational data in the accompanying two papers. The effect of universal moderate dietary salt reduction on mortality from stroke and ischaemic heart disease would be substantial--larger, indeed, than could be achieved by fully implementing recommended policy for treating high blood pressure with drugs. However, reduction also in the amount of salt added to processed foods would lower blood pressure by at least twice as much and prevent some 75,000 [corrected] deaths a year in Britain as well as much disability.